⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ

WEEK 03

Collections, loops and totals

បណ្តុំផ្ទាំង រង្វិលជុំ និងផលបូក

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WEEK 03

This week: learn, practise, build

សប្តាហ៍នេះ រៀន អនុវត្ត និងបង្កើត

01 · Python

Collections, loops and totals
Understand the idea and trace examples.

02 · Practice

Practise the same Python concept.
Predict, change, diagnose and explain.

03 · Project

Multi-batch aggregation
Build, discuss and test with a peer.

One weekly component moves the final project forward.

ផ្នែកមួយរាល់សប្តាហ៍ជួយឱ្យគម្រោងចុងក្រោយរីកចម្រើន។

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Python foundations

មូលដ្ឋាន Python

Collections, loops and totals
Concept → trace a small example → explain its behavior

អានបណ្តុំនិយម ហើយហូតផ្តុំរួមជុំ។

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Lists, records and accumulation

List កំណត់ត្រា និងការហូរសន្សំ

A list holds records

[0] selects the first record.
["points"] selects its field.

The total has a lifetime

Reset before the loop.
Update inside; report after.

A count is not a sum

len(records) counts rows.
Adding points measures total points.

Trace a small list, then write a loop that also handles empty input.

តាមដាន list ហើយ សរសេរ loop ដែលដោះស្រាយទាំងទទេផង។

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Store many values, then process each one

រក្សាតម្លៃច្រើន ធ្វើដំណើរការម្តងមួយ

List: an ordered collection

Use an index to select one item. A list may contain dictionaries.

Dictionary: named fields

Read a value by its key, such as row["count"]. Keys describe the data.

Loop: repeat a block

for visits each item. Start a total once, then add each contribution inside the loop.

Understand the behavior before changing the example.

យល់ឥរិយាបថក្នុងមុនកែឧទាហរណ៍។

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A loop keeps its running state

រង្វិលជុំក្បាស្តីនភាពរត់ការមុន

1

Before the loop

total = 0
Initialize once

2

For each item

Read fields
Calculate and add

3

After the loop

Report the final total
Empty list leaves zero

Read the diagram, then trace the same steps in code.

អានដ្យាក្រាម រួចតាមដានជំហានដដែលក្នុងកូដ។

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Syntax to read and reuse

វាក្យសម្ពន្ធសម្រាប់អាន និងប្រើឡើងវិញ

Python / syntax	Meaning	When to use it
[item, item]	List: ordered items	Store several records
{"key": value}	Dictionary: named fields	Describe one record
for item in items:	Visit each item	Process a collection
total += value	Add to the running total	Accumulate contributions

Python Basics · sections 05, 06 · use only when you need a lookup.

Python Basics ផ្នែក 05, 06៖ មើលយោងតែពេលចាំបាច់។

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Read fields and accumulate a total

អានវាលឱ្យបាន និងបូកសរុប

```
records = [{"name": "A", "count": 2},
           {"name": "B", "count": 3}]
print(records[0]["name"])
total = 0
for row in records:
    total += row["count"]
    print(row["name"], total)
print("total", total)
```

1. Expected: A; A 2; B 5; total 5.
2. The accumulator starts before the loop.
3. Try an empty list: what still runs?

Predict → execute → explain the value at each step.

ព្យាករ → ដំណើរការ → ពន្យល់តម្លៃនៅជំហានមួយៗ។

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Lists contain items; dictionaries contain fields

list ផ្ទុកធាតុ; dictionary ផ្ទុកវាលឈ្មោះ

1. **List** — Ordered items. A position selects one item; indexing starts at zero.
2. **Dictionary** — Named key/value pairs. A key selects one field.
3. **Value** — The selected value can be used in an expression or another statement.

A list position and a dictionary key select different things.

ទីតាំងក្នុង list និង key ក្នុង dictionary ជ្រើសរបស់ខុសគ្នា។

Read a dictionary value by key

អានតម្លៃdictionary តាម key

```
01 line = {"batch_id": "T01",
02         "portions_per_container": 3, "containers": 4}
03 print(line["batch_id"])
04 print(line["containers"])
05 planned = line["portions_per_container"]
06 planned *= line["containers"]
07 print("Planned portions:", planned)
08 assert planned == 12
```

1. Predict the ID, count and labelled portion estimate
2. Keys retrieve named values
3. A wrong key spelling raises KeyError

The estimate combines two fields from the same record.

ការប៉ាន់ប្រមាណផ្សំលេងផ្ដុំតែពីរតែមួយ។

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for: visit each item in a list

for៖ ដើរកាត់ធាតុមួយៗក្នុង list

```
01 lines = [
02     {"batch_id": "T01", "containers": 4},
03     {"batch_id": "T02", "containers": 2},
04 ]
05 for line in lines:
06     print(line["batch_id"], line["containers"])
07 print("Lines:", len(lines))
08 assert len(lines) == 2
```

1. Write the output in execution order.
2. The indented print runs for each item.
3. The final print runs after the loop.

The loop body runs once per item; len() counts items.

រូបរៀបជុំគ្នាក្នុងមួយធាតុ; len() រាប់ធាតុ។

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Trace the accumulator after each contribution

តាមដានផលហូរក្រោយបន្ថែមនីមួយៗ

```python
contributions = [12, 8]
total = 0
for contribution in contributions:
    before = total
    total += contribution
    print(before, "+", contribution, "=", total)
assert total == 20
```

1. Trace the first contribution from the initial zero
2. Use that running total for the second contribution
3. The previous total is part of the next calculation

Initialize once, add each contribution, then report.

កំណត់តម្លៃដើម្បី ហូតនីមួយៗ ជូនរាយការណ៍។

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An empty list: zero loop iterations

list ទទេរ មិនធ្វើដំណើរការ

```
01 contributions = []
02 planned_portions = 0
03 for contribution in contributions:
04     planned_portions += contribution
05 print("Planned portions:", planned_portions)
06 assert planned_portions == 0
```

1. How often does the body execute?
2. Which value remains afterward?
3. Initialize before the loop, even for [].

An empty list skips the loop body and keeps the initial value.

list ទទេរំលងតួរមិនធ្វើរហើយរក្សាតម្លៃដើម។

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Hands-on practice

លំហាត់អនុវត្តផ្ទាល់

Week_03.ipynb · Session 2
Predict → run → change → explain

ព្យាករ រត់ ប្តូរយ៉ាង និងពន្យល់លទ្ធផល។

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Your turn: practise the same Python skills

ដល់វេនអ្នក អនុវត្តជំនាញ Python ដដែល

P1–P2 · Easy

Use the approach hints. Write the code yourself in a blank cell.

P3 · Core build

Implement a complete small program. Write expected results and assertions.

P4–P5 · Optional

Apply the skill to a new case, then attempt a harder challenge if time permits.

Write your own code, predict the result, then test and explain it.

សរសេរកូដផ្ទាល់ ព្យាករលទ្ធផល ឆ្លុះសាក និងពន្យល់។

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Check your work before asking for help

ធ្វើការងារមុនសុំជួយ

A complete implementation

Write the whole operation. Do not stop at one changed character.

Independent expected results

Calculate a normal case and an edge case before running your program.

A peer can explain it

Swap driver and reviewer. Explain one line and one test.

Read list/dictionary values and trace an accumulator, including empty input.

អាន list/dictionary និងតាមដានផលបូក មួទាំងឱ្យទទេ។

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Project studio

វគ្គសិក្សារយៈពេលខ្លី

Multi-batch aggregation
Short design briefing → most time for your work

ហូចំនួនធុង និងចំណែកតាមផែនការណែនាំអាហារថ្មី។

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The people’s task this week

កិច្ចការរបស់អ្នកច្រើនាសប្តាហ៍នេះ

1

Someone needs help

A pickup can contain more than one kind of food.

2

This week, you will

Read each food line and add its containers and planned portions.

3

A useful result

The coordinator can read the totals without mixing lines, containers and portions.

Use the supplied starting files. Focus on one useful result and check it.

ប្រើកសារចាប់ផ្តើមដែលផ្តល់។ ផ្តោតលើលទ្ធផលមានប្រយោជន៍មួយ ហើយធ្វើតេស្ត។

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Test a split without changing the underlying plan

សាកល្បងបំបែកដោយមិនប្តូរផែនការ

1

Original

One four-container T01 line

2

Transform

Two two-container T01 lines with the same declaration

3

Predict

One extra line; unchanged container and portion totals

4

Check

Compare both representations with independent expectations

Line count may change while containers and planned portions stay equal.

ចំនួនជួរអាចប្តូរតែធុង និងចំណែកតាមផែនការនៅដដែល។

Your implementation TODO

ការងារដែលអ្នកត្រូវសរសេរក្រុម

Supplied starting point

The notebook supplies BREAD-A and RICE-A line dictionaries.

You implement / do

Read each line, multiply containers by unit size, and accumulate container and portion totals outside/inside the loop correctly.

Check the evidence

Starter: 2 lines, 5 containers, 16 portions. Empty list → zeros. Split the bread line: total portions must stay 16.

Explain one field choice and trace one contribution into the total.

ពន្យល់ការជ្រើសវាលមួយ និងតាមដានតម្លៃមួយចូលផលបូក។

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Show the component working

បង្ហាញថាផ្នែកនេះដំណើរការ

Step	What to show
1 · Locate	Identify the input, this component and its next user.
2 · Implement	Loop over lines; accumulate containers and portions.
3 · Check	Starter 2/5/16; empty → zeros; split line preserves total.
4 · Explain	Show your code, its actual result and why it behaves that way.

Explain one field choice and trace one contribution into the total.

ពន្យល់ការជ្រើសវាលមួយ និងតាមដានតង្ហែយចូលផលបូក។

Further reading in English

ឯកសារអានបន្ថែមជាភាសាអង់គ្លេស

1 Python tutorial: data structures

https://docs.python.org/3/tutorial/datastructures.html

2 Python tutorial: for statements

https://docs.python.org/3/tutorial/controlflow.html#for-statements

Read one relevant section, then bring an example and a question.

អានផ្នែកដែលពាក់ព័ន្ធមួយ ហើយយកឧទាហរណ៍មួយ និងសំណួរមួយមកវិញ។